Lavish Avinash

Software Engineer | Backend Systems Architect | AI & Agentic Systems Developer | Product Builder
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PROFESSIONAL SUMMARY

Software engineer (B.E. Information Science, 2027) focused on backend architecture and AI developer tooling. Architected the MandiMangoes commerce backend on Node.js, Express and PostgreSQL 16, with transactional inventory, idempotent Razorpay payments, role-based access and 900+ automated tests. Builds MCP-based code impact analysis for AI coding agents and Transformer-based demand forecasting. Winner, DevDays Bengaluru Build OpenAI Hackathon.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL
Backend: Node.js, Express.js, FastAPI, REST API design, Authentication & authorisation (JWT, RBAC, TOTP), Webhooks & idempotency
Databases: PostgreSQL, MySQL, SQLite, Schema design & migrations, Transactions & locking
Frontend: React, Next.js, HTML, CSS
AI / ML: PyTorch, Transformers (time series), Scikit-learn, Pandas, NumPy, Feature engineering, Model evaluation
AI agents: Model Context Protocol (MCP), AI agent integrations, Dependency graph analysis
Infrastructure & tooling: Git, GitHub Actions (CI/CD), Vercel, Neon, Docker, Playwright, Linux, Postman

PROFESSIONAL EXPERIENCE

Founder & Backend Systems Architect Present
MandiMangoes, Bengaluru, India

- Architected the MandiMangoes backend as a modular monolith on Node.js, Express and PostgreSQL 16, serving the storefront API, admin API and Razorpay payments from one service and one database.
- Designed a transactional stock engine in which time-limited holds, payment confirmation and FEFO batch allocation commit in a single PostgreSQL transaction, using row-level locks (FOR UPDATE, SKIP LOCKED) to prevent overselling.
- Built idempotent payment handling: HMAC-SHA256 webhook verification with timing-safe comparison, replay-safe payment outcomes, idempotency keys on refunds and automatic refunds of duplicate or wrong-amount payments.
- Implemented four-role access control (manager, operator, packer, QC) with TOTP two-step sign-in for staff and an activity log recording every staff change.
- Covered the backend with 900+ automated test cases and the React admin panel with Playwright end-to-end, axe accessibility and API-contract tests.

Software Engineering Intern Present
VMR Cloud Solutions, Bengaluru, India

Digital Content Producer & Brand Communications Associate Present
Yashoda Elevators, Bengaluru, India

- Conceptualise promotional campaigns and brand stories for the company's digital channels.

PROJECTS

AMU: Anti Messup Agent
Model Context Protocol (MCP), Dependency graphs, Static code analysis

- Built an AI-agent-oriented code impact analysis system that assesses the downstream effects of a proposed repository change before an agent makes it.
- Designed a repository dependency graph tracing direct and transitive relationships between functions and modules, exposed to AI coding agents as MCP tools with source-backed dependency paths.
- Surfaced related tests and documentation for targeted validation, with coverage and uncertainty information so agents can tell evidence from gaps.

Aurora
Python, PyTorch, FastAPI, Next.js 14, MySQL

- Architected an end-to-end multi-SKU demand forecasting pipeline around a Transformer time-series model in PyTorch (30-day input, 5 features, 7-day horizon).
- Turned forecasts into automated safety stock and reorder point calculations served through a FastAPI inference API and a Next.js 14 interface.

EDUCATION

Bachelor of Engineering, Information Science and Engineering Expected 2027
CMR Institute of Technology, Bengaluru | CGPA: 8.22/10

AWARDS & ACHIEVEMENTS

- Winner, DevDays Bengaluru Build OpenAI Hackathon. Top 10 of approximately 1,000 participating teams.
- Shortlisted, ISRO Robotics Challenge (IRoC-U). Autonomous UAV navigation and obstacle detection.
- First Place, Flex It Fitness Competition. Cash prize for outstanding performance.